

Marrimanu Charan Teja

Master's Student | Data Engineer & AI Researcher | Indian | 15th August 2002

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PROFILE SUMMARY

Results-driven Data Engineer and AI Researcher with expertise in **distributed computing (Dask)**, **geospatial analytics**, **machine learning**, and **deep learning**. Currently pursuing a Master's in CS & Engineering at The University of Aizu, Japan, specializing in traffic congestion prediction, one-class classification of raster images, and time-series forecasting. Contributor to open-source libraries **geoAnalytics** and **PAMI**. Experienced in deploying ML pipelines on heterogeneous GPU/CPU clusters (NVIDIA Jetson Orin + AMD EPYC). Invited speaker at ETLTC2026 and Best Poster Award winner.

TECHNICAL SKILLS

Languages: Python (Expert) | R (Intermediate) | C/C++ (Proficient) | SQL (Proficient) | Java | JavaScript

ML/AI: Transformers, Machine Learning, Deep Learning, NLP, Fuzzy Systems, One-Class Classification, Multi-Task Learning

Data Engineering: Dask (Distributed Computing), Numba (Parallel Computing), PostgreSQL, ETL Pipelines, Time-Series Forecasting, XGBoost, Random Forest

Geospatial: GDAL, QGIS, Raster Mining, geoAnalytics library, CSV-to-Raster conversion

Infrastructure & Tools: Linux (Ubuntu), NVIDIA Jetson Orin NX (CUDA/PyTorch), Proxmox/JupyterHub, macOS, Windows

Version Control & Others: Git/GitHub, PAMI, Power BI, Tkinter, PyInstaller, Custom GPT APIs

EDUCATION

The University of Aizu, Japan | Oct 2024 – Present

Master of Science in Computer Science & Engineering | **CGPA: 4.0 / 4.0**

- Research: A Probabilistic Attention-Calibrated Forecasting Framework for Route Recommendation System
- Supervisor: Prof. RAGE Uday Kiran | Lab: Intelligent Transportation Lab

Madanapalle Institute of Technology & Science, India | Nov 2021 – Jun 2024

B.Tech in Artificial Intelligence | **CGPA: 8.9 / 10**

- Best Thesis Award — Knowledge Discovery in Japan Traffic Congestion Data
- Hosted RAIDS Symposium on AI & Data Science with industry and academia

KEY PROJECTS

A Real-Time Route Recommendation System for Autonomous Vehicles Using a Probabilistic Attention-Calibrated Forecasting Framework (Master's Thesis)

- **Encoder-Decoder Bi-LSTM** with 8-head multi-head attention for probabilistic traffic forecasting (dual heads for μ and σ^2 via Negative Log Likelihood loss)
- Achieves **95% PICP** and **0.743 calibration score** on sparse data (105k timesteps, 90% zeros)
- **Multi-criteria route scoring** integrating real-time traffic predictions + rainfall data – outperformed Google Maps
- End-to-end pipeline: Yahoo Weather & Soramame APIs → PostgreSQL → Dask distributed processing → route comparison across Fukushima road network
- **Tech Stack:** PyTorch, CUDA, Bi-LSTM, Attention, Dask, PostgreSQL, REST APIs, Python

geoAnalytics – Open Source Geospatial Python Library

- Built a standalone **macOS DMG application** making one-class classification accessible to non-programmers
- **One-Class Classification** – Correlation & FuzzyTSC classifiers for anomaly detection in raster images
- **Additional capabilities:** Clustering, CSV-to-Raster conversion, Spatial interpolation (imputation), Normalization, Fuzzy geo-referenced pattern mining, QGIS plugins for visualization
- **Tech Stack:** Python, GDAL, Tkinter, PyInstaller, QGIS, Scipy

PUBLICATIONS

[C1] Charan Teja Marrimanu, Arjun Chakravarthi Pogaku. — [A Novel Multi-Task Learning Framework for Predicting Traffic Congestion](#). Big Data & AI Conference, 2024.

[C2] Vanitha Kattumuri, Rage Uday Kiran, Charan Teja Marrimanu, Yoshiko Ogawa, Ohtake Makiko. — [geoAnalytics: An Open Source Python Library for Geospatial Data Analytics](#). IEEE Space Conference, 2025.

RESEARCH & WORK EXPERIENCE

Research Assistant — The University of Aizu | Oct 2024 – Present

Intelligent Transportation Lab & geoAnalytics Team

- **Built distributed ML pipeline** for PM2.5 time-series forecasting using Dask across heterogeneous cluster (Jetson Orin aarch64 + AMD EPYC x86_64, 128 CPUs)
- **Developed route recommendation system** integrating real-time traffic & rainfall data – algorithm route scored **0.1897** (safer, 4.59mm rainfall) vs Google Maps **0.1937**
- Engineered **geoAnalytics** Python library modules: oneClassClassifiers (Correlation, FuzzyTSC), CSV-to-Raster conversion, QGIS plugins for raster data
- **Benchmarked ARIMA, ETS, LR, RF, KNN** on PM2.5 data via Dask – **GRU outperformed all** models on traffic forecasting
- **Contributed test cases & documentation** to PAMI open-source pattern mining library (Apriori, PFECLAT, ECLATbitset)
- **Processed Moon/Mars multi-band TIFF spectral data** – applied spatial interpolation (scipy.griddata) for imputation
- Researching distributed mining of **Partial Periodic Patterns** in large-scale time-series data using Dask across multi-node clusters
- **Developing fuzzy geo-referenced pattern mining** algorithms to discover spatial co-occurrence patterns in PM2.5 and environmental datasets
- **Investigating periodic pattern discovery** in weather and traffic sensor data for predictive environmental analytics
- **Applied fuzzy geo-referenced pattern mining** to uncover spatial co-occurrence trends in PM2.5, rainfall, and road congestion datasets
- **Implemented optimized one class classifiers** using MaxNorm, Manhattan, and Hausdorff distances in single-threaded, parallel, and CUDA modes for large-scale raster classification
- **Conducted Average Distance and Cluster Overlap evaluations** for similarity-based one-class classification benchmarking

Teaching Assistant — The University of Aizu | Oct 2024 – Present

Undergraduate Courses

- Core CS: C++, Data Structures & Algorithms, Software Engineering, Web Engineering
 - Data & AI: Big Data Analytics, Intro to Data Science (Python), Artificial Intelligence
 - Mathematics: Calculus, Dynamics
- *Responsibilities:* Conduct lab sessions, grade assignments, hold office hours, assist with project development

Student Intern — The University of Aizu, Japan | Jan 2024 – Mar 2024

Knowledge Discovery in Traffic Congestion Data

- Performed knowledge discovery in real-time Japan traffic congestion data and documented PAMI library

Virtual Intern — MathWorks (AICTE), India | May 2023 – Sep 2023

AI Domain Application

- Applied AI in domain-specific technical intelligence under AICTE-sponsored MathWorks program

HONORS & AWARDS

- 🏆 Best Poster Award — 4th Graduate Forum 2025, The University of Aizu
- 🏆 Best Thesis Award — MITS University 2024 (Japan Traffic Congestion Internship)

ACADEMIC & PROFESSIONAL ACTIVITIES

- Invited Speaker — ETLTC2025 Conference: "Discovering Hidden Knowledge in Raster Images"
- Technical Volunteer & Question-Panel Member — ETLTC Conference Series
- Poster Presentations — Postgraduate Forum 2024 & 2025, The University of Aizu (UBIC)
- Reviewer — International conferences in computational intelligence and data engineering
- Volunteer — SAISUA Events, Contributor to 2025 Fukushima World Wide Learning HS Summit
- Cultural Ambassador — Teaching Kho-Kho, Lagori & Bollywood dance to Aizuwakamatsu school children

LANGUAGES

English (Fluent) | **Japanese** (Conversational) | **Hindi, Telugu, Kannada** (Native)